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3 Distance protection

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3.1 Setup of the network model

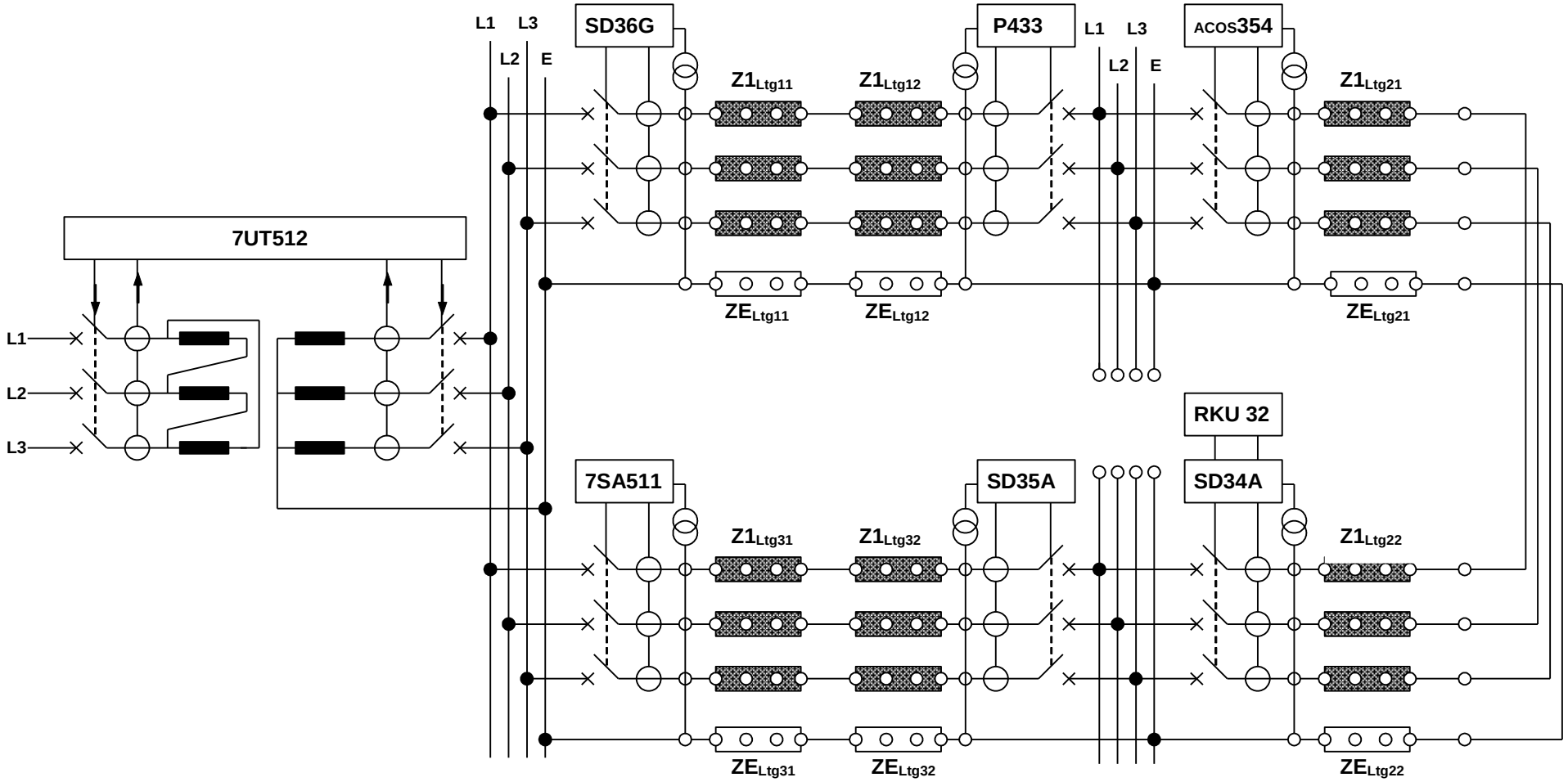


Figure 3.1.1 Setup of the grid model. Ring grid

Darmstadt, 2021

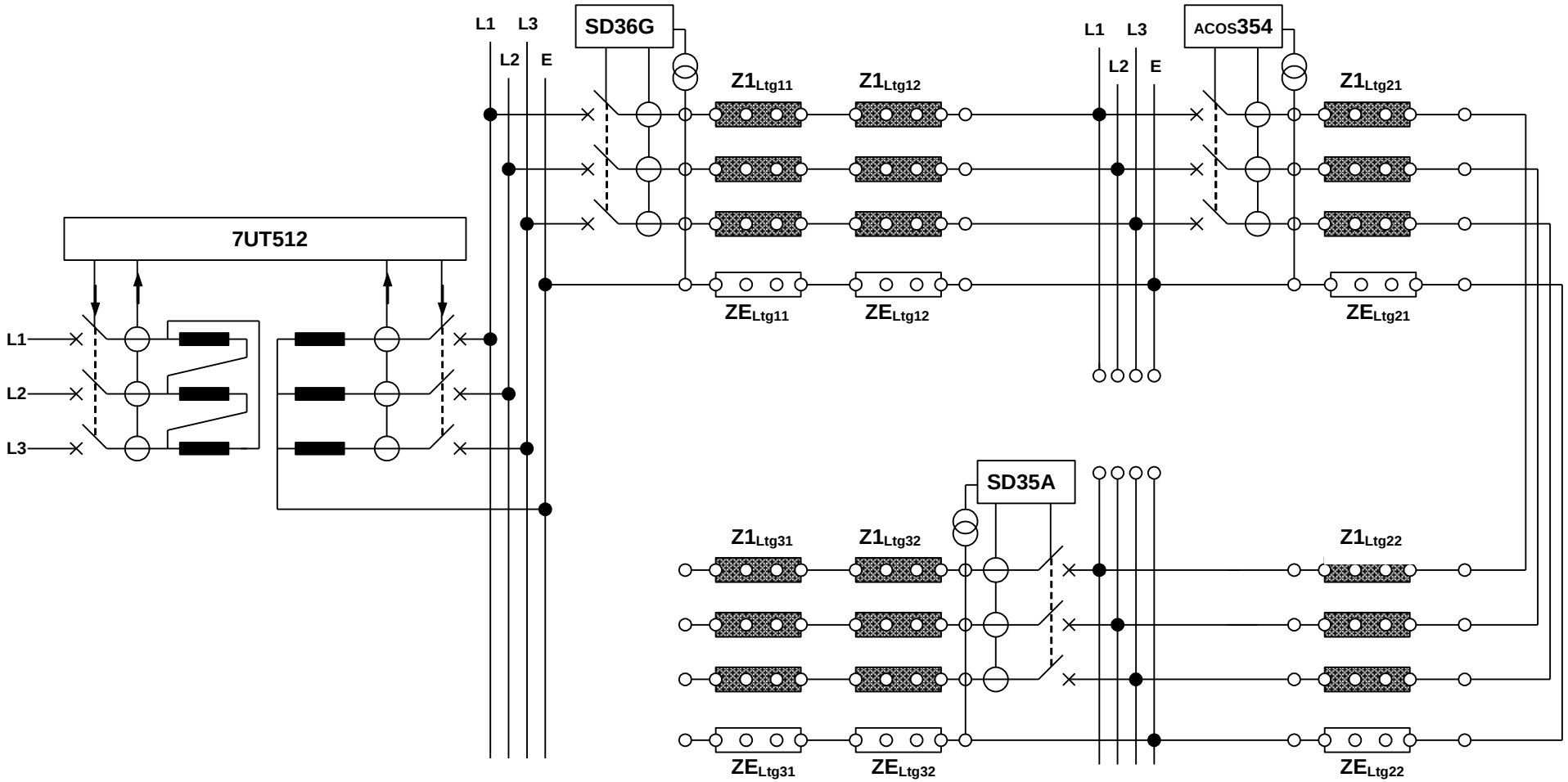


Figure 3.1.2 Setup of the grid model. Type A radial grid

Darmstadt, 2021

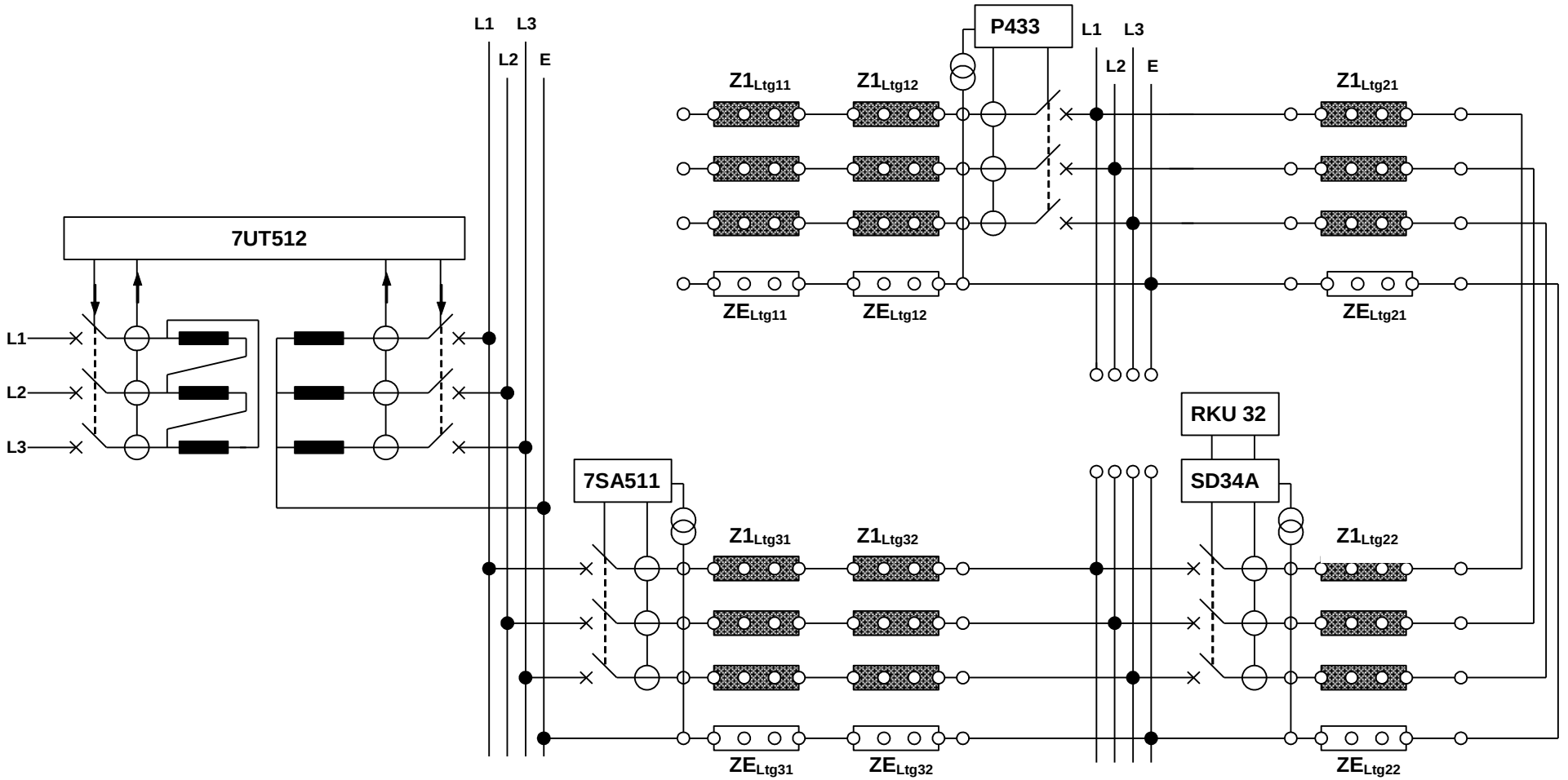


Figure 3.1.3 Setup of the grid model. Radiating grid type B

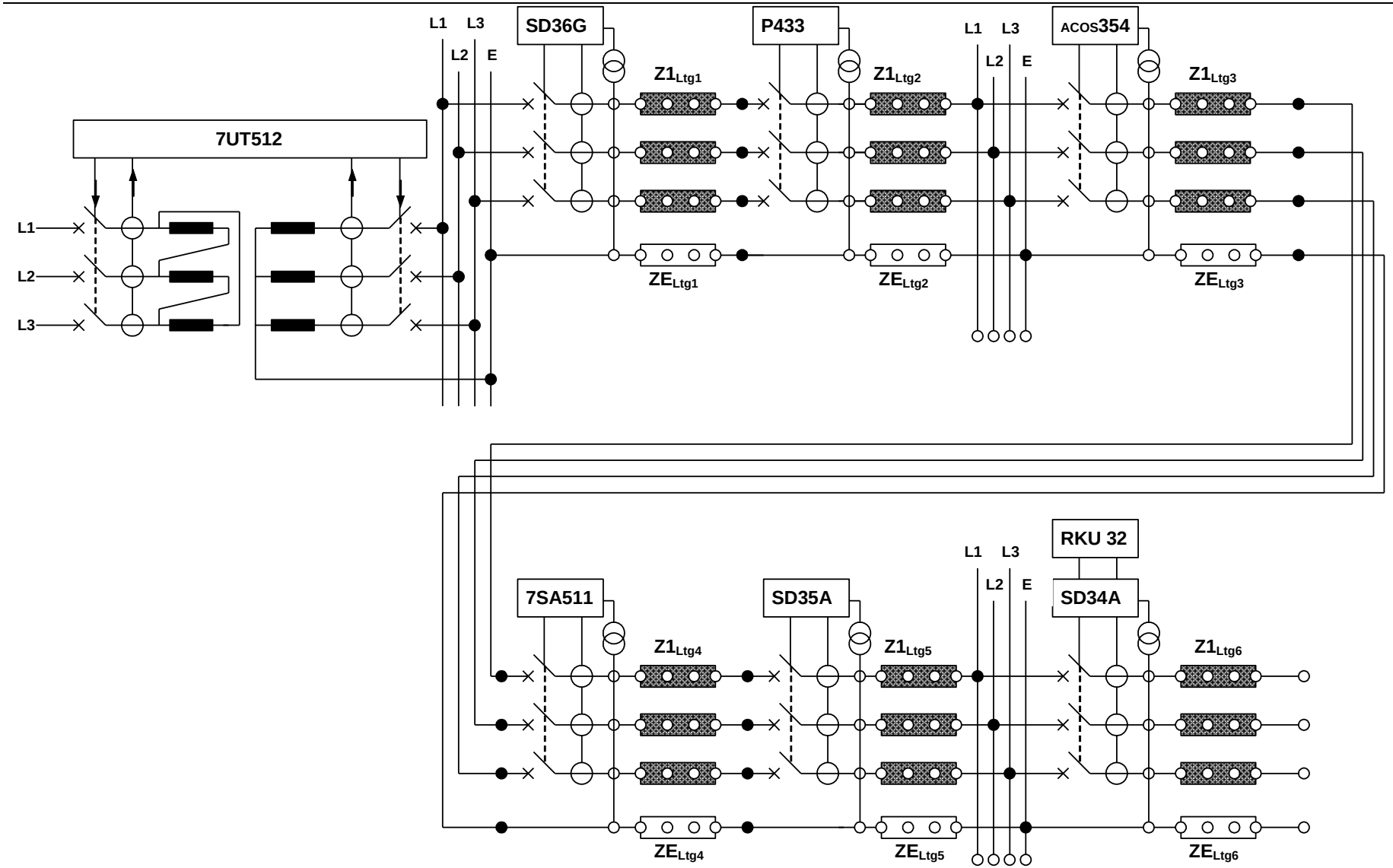


Figure 3.1.4 Structure of the grid model. Radiating grid type C

Darmstadt, 2021

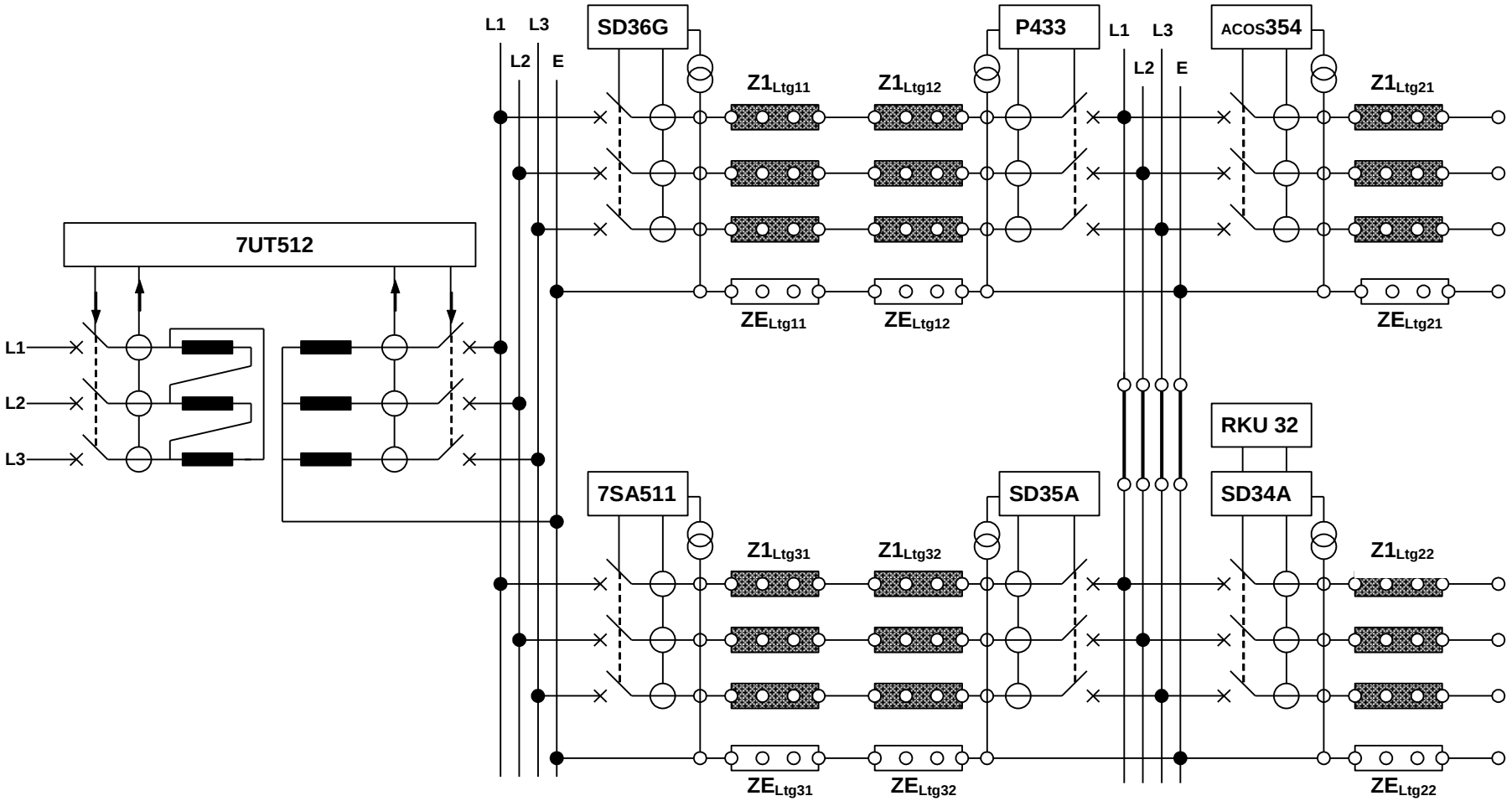


Figure 3.1.5 Structure of the network model. Parallel line

3.2 Distance protection devices

3.2.1 General

Protective devices in MV and HV networks.

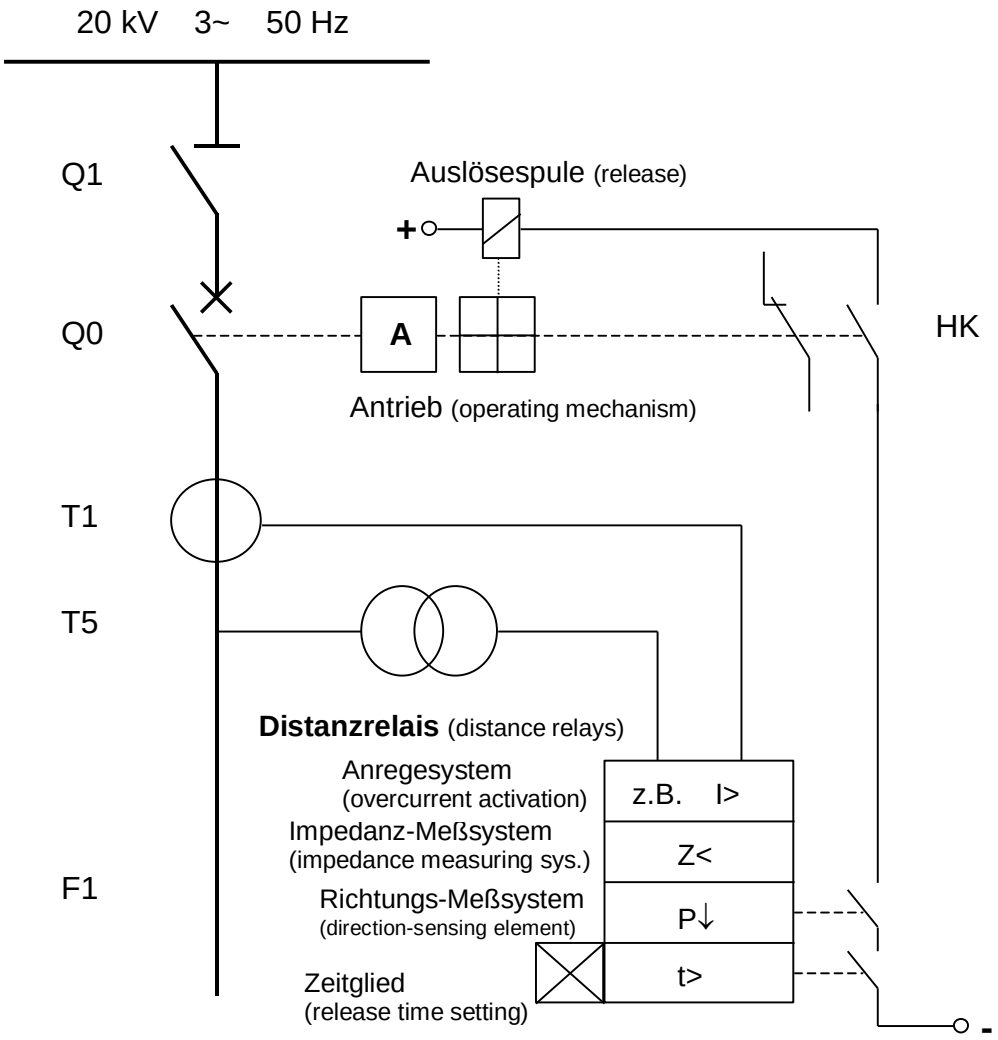
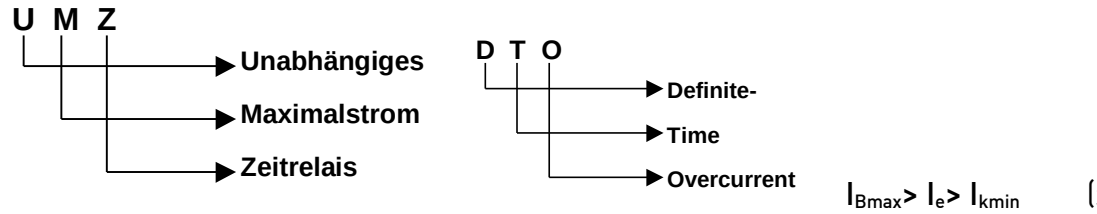
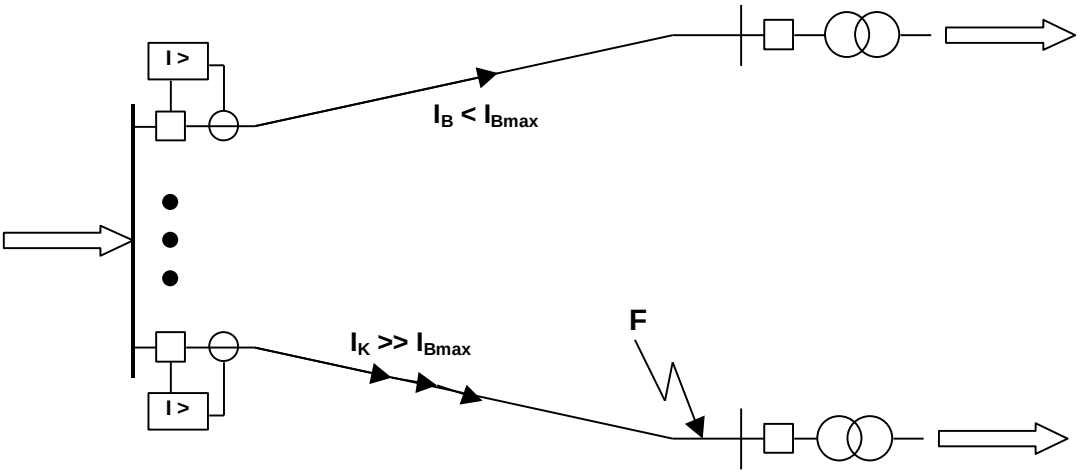


Figure 3.2.1 Basic structure of relay protection:
Q1 – Disconnector
Q0 – Circuit breaker
T1 – Current transformer
T5 – Voltage transformer
F1 – Protection device
HK – Auxiliary switches

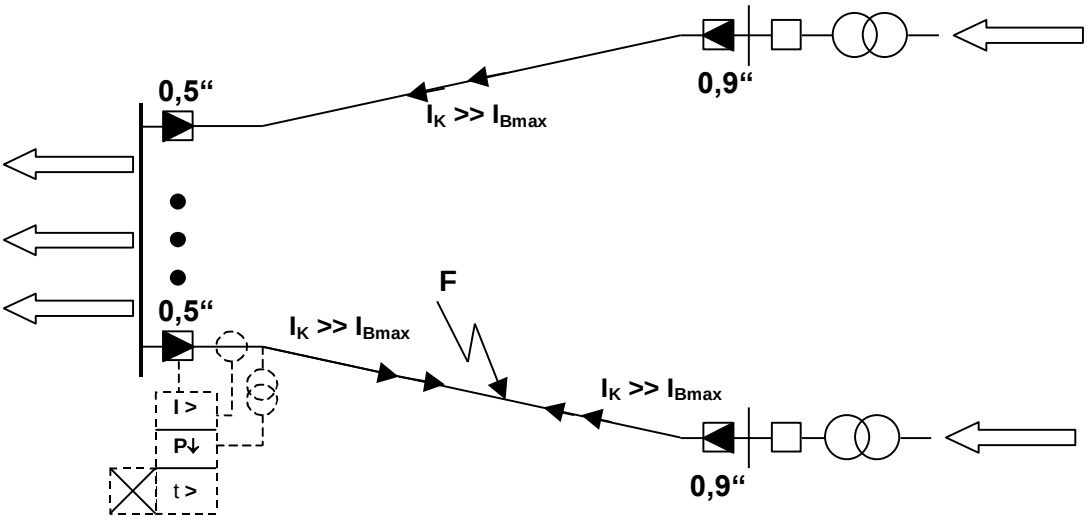
Undirected and directed UMZ relays (undirectional and directional DTO relays)



$$I_{Bmax} > I_e > I_{kmin} \qquad (3.2.1)$$



a)



b)

Figure 3.2.2 Single-ended lines with unidirectional DTO relays (a) and double-ended lines with directional UMZ relays (b);
I_k – Short-circuit current, I_B – Operating current

Selectivity through time staggering. Time staggering: **0.3 s ... 0.5 s**

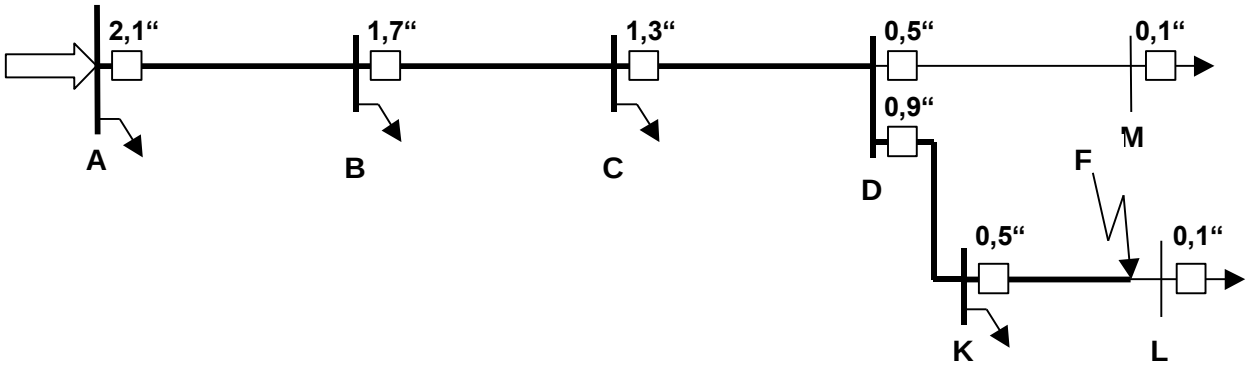


Figure 3.2.3 Time delay in single-ended lines with non-directional UMZ relays

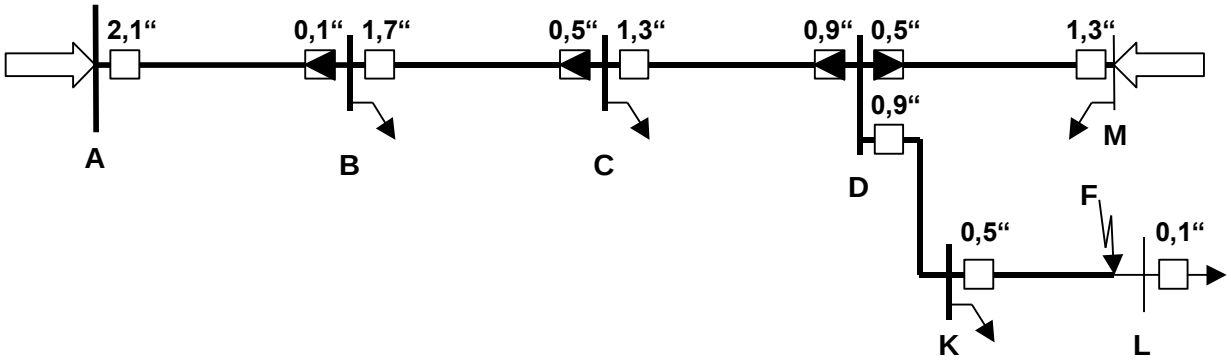
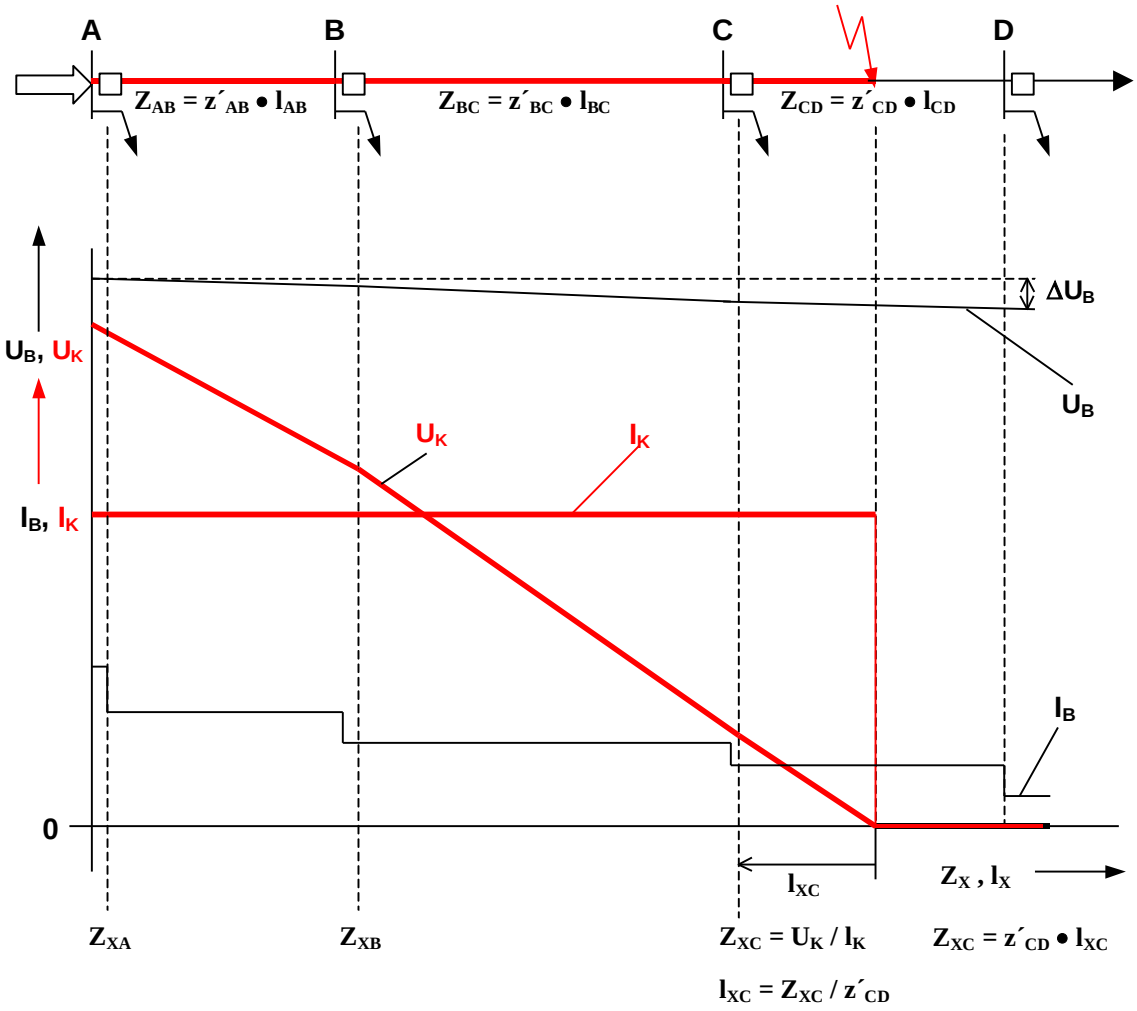


Figure 3.2.4 Time delay in double-fed lines with directional UMZ relays

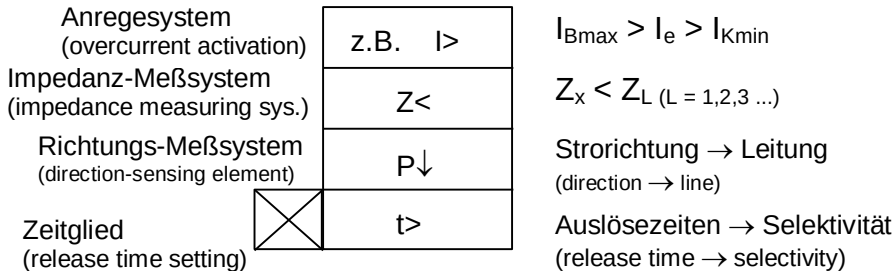
Rules for assigning directional elements in double-fed lines:

1. Of the two UMZ relays adjacent to a busbar, only the relay with the shorter trip time
2. If several UMZ relays adjacent to a busbar are set to the same trigger times, all UMZ relays receive a directional element that is located in on lines fed from two sides.

Functional principle of distance protection



a)

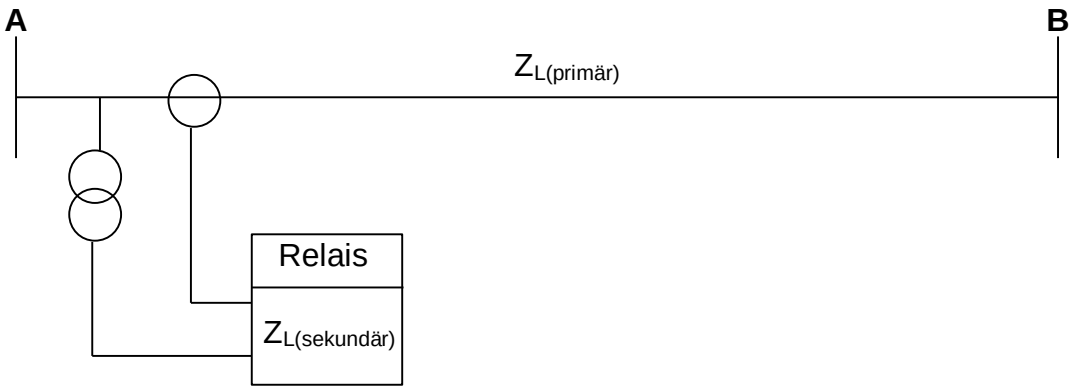


b)

Figure 3.2.5 Functional principle of distance protection. Current and voltage conditions in the radial network in an undisturbed state and during a short circuit (a) as well as a block diagram of a distance protection (b)

Darmstadt, 2021

Transformation of the step impedance



$$Z_{L(sekundär)} = \frac{U_{sekundär}}{I_{sekundär}} = \frac{U_{primär} \cdot \frac{1}{k_{NU}}}{I_{primär} \cdot \frac{1}{k_{NI}}} = \frac{U_{primär}}{I_{primär}} \cdot \frac{k_{NI}}{k_{NU}} = Z_{L(primär)} \cdot \frac{k_{NI}}{k_{NU}}$$

$$Z_{L(sekundär)} = Z_{L(primär)} \cdot \frac{k_{NI}}{k_{NU}} \qquad (3.2.2)$$

Setting the distance protection. Relay coordination principles

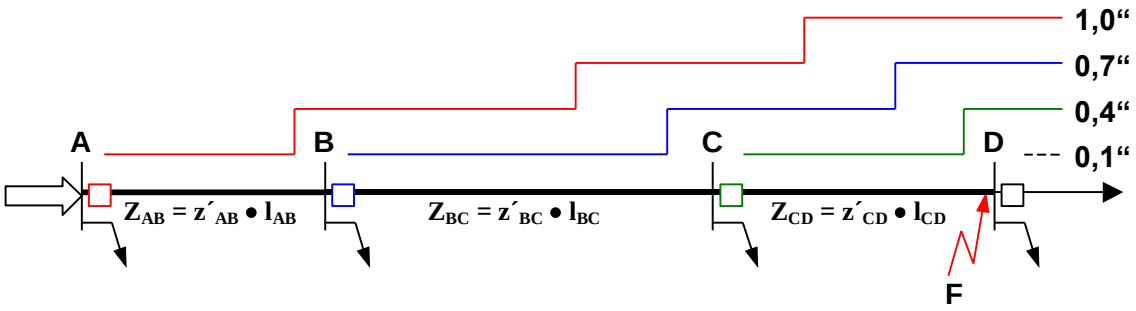


Figure 3.2.6 Time relay coordination plan for distance protection devices in radial networks

Setting for distance relay **A:**
1st stage (fast stage) for 90% of the distance AB with trip time t_A < 0.1 s
$Z_X < Z_{1.St} = 0{,}9 \cdot Z_{AB}$
2nd stage for the remaining 10% of the distance AB and as reserve protection for part of the distance BC
$Z_{1.St} < Z_X < Z_{2.St} = 0{,}8\,(Z_{AB} + 0{,}9 \cdot Z_{BC})$ with a delay of 0.3 s ... 0.5 s, e.g., T_1 = 0.4 s
3rd stage as reserve protection with a delay of a further 0.3 s ... 0.5 s, e.g., T_2 = 0.7 s
$Z_{2.St} < Z_X < Z_{3.St} = 0{,}8\,[Z_{AB} + 0.8\,(Z_{BC} + 0{,}9 \cdot Z_{CD}]$

Darmstadt, 2021

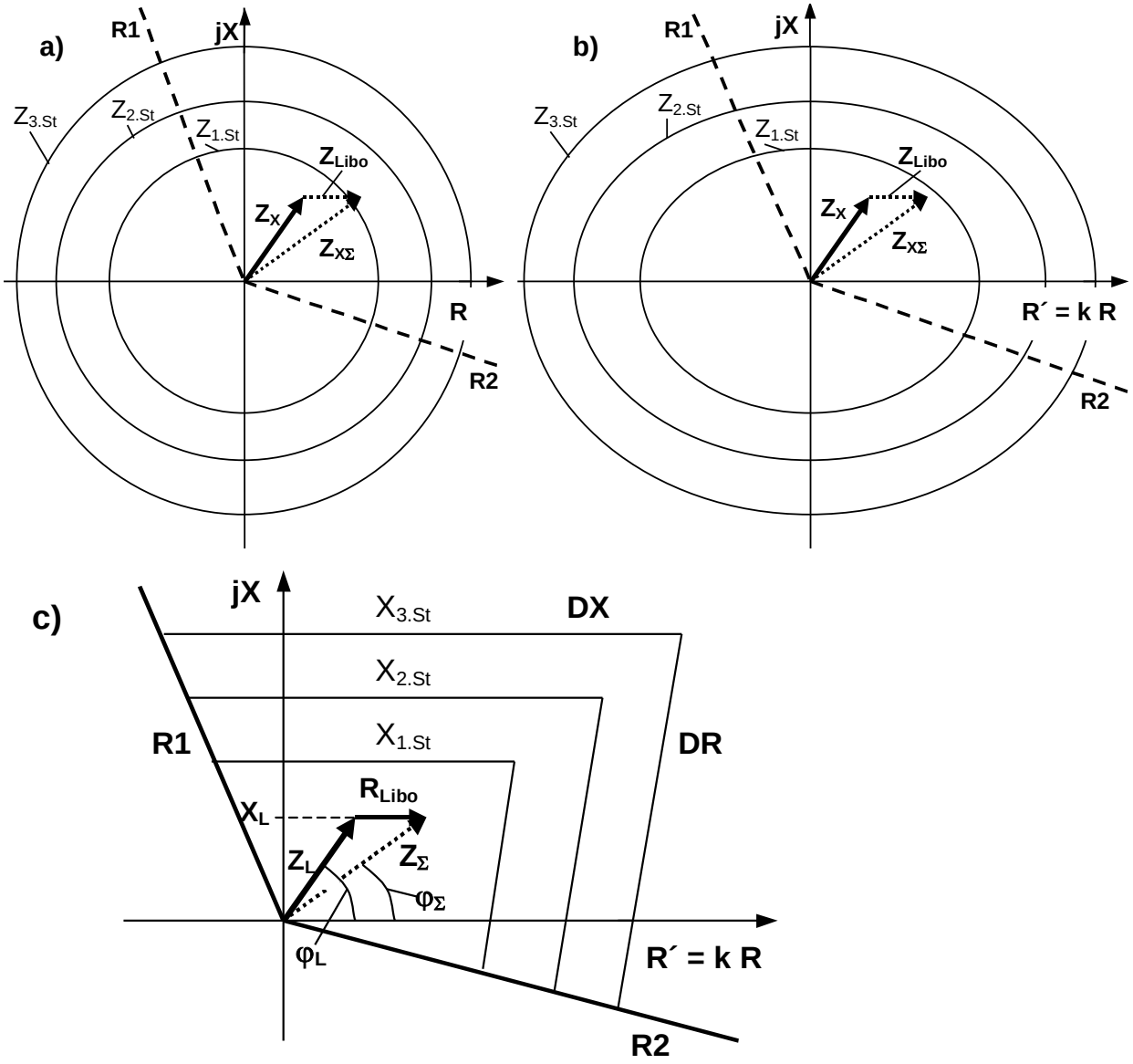


Figure 3.2.7 Selected trip characteristics of distance protection devices:

a) Circular characteristics (impedance relay – circular char.)
b) Elliptical characteristics (impedance relay – ellipsoidal char.)
c) Polygonal characteristics, DX, DR – distance characteristics; (quadrilateral relay)

$$\underline{Z}_\Sigma = R_\Sigma + jX_\Sigma = Z_\Sigma \cdot e^{j\varphi_\Sigma} = Z_\Sigma \cdot \cos\varphi_\Sigma + j\, Z_\Sigma \cdot \sin\varphi_\Sigma$$

$R_\Sigma = Z_\Sigma \cdot \cos\varphi_\Sigma \quad R_\Sigma = R_L + R_{Libo}$, varies depending on $R_{(Libo)}$

$X_\Sigma = Z_\Sigma \cdot \sin\varphi_\Sigma = Z_L \cdot \sin\varphi_L = X_L$, constant, independent of $R_{(Libo)}$

3.2.2 Distance relays used in the network model

The following protection relays are currently used in the FHD protection technology laboratory:

- Electromechanical relays (SD34A, RKU 32)
- Electronic relay SD35A
- Digital relays (SD 36G, SIPROTEC 7SA511, MICOM P433, ACOS 354)

These are distance relays that are very common in practice. They range from old electromechanical relays that were built in the 1960s and are still in operation (but are no longer being installed) to the most modern digital relays that are state of the art today.

3.3 Demonstration in the protection technology laboratory

In the protection technology laboratory, a lecturer demonstrates the following experiments:

- Setting the SD 34A distance relay and testing it in a radial network (network model, radial network type B)

- Setting the RKU relay and testing its interaction with the SD 34A distance relay (network model, radial network type B)

- Setting the SIPROTEC 7SA511 distance relay and testing it in a ring network (network model, ring network)

Setting and information output using a PC

3.3.1 Tests with the SD 34A distance relay

3.3.1.1 Setting the SD 34A distance relay

Current transformer: 5A/5A Voltage transformer with k_{NU}=1

A) Excitation system
Overcurrent excitation: I> = 5A
B1) Impedance measurement – impedance stages
Circuit characteristic curve

$$Z_{1.St} = 0.9 \cdot 3.44\Omega \cdot \frac{k_{NI}=1}{k_{NU}=1} \cdot \frac{1}{C=4} = 0.77\Omega$$

$$Z_{2.St} = 0.8\,(3.44\Omega + 0.9 \cdot 3.44\Omega) \cdot \frac{k_{NI}=1}{k_{NU}=1} \cdot \frac{1}{C=4} = 1.31\Omega$$

$$Z_{3.St} = \max. = 2.5\Omega$$

Elliptical characteristic curve

$$X_{1.St} = 0.9 \cdot (1.55\Omega + 1.57\Omega) \cdot (m=1.95) \cdot \frac{k_{NI}=1}{k_{NU}=1} \cdot \frac{1}{C=4} = 1.37\,\Omega$$

$$X_{2.St} = 0.8\left[3.12\Omega + 0.9 \cdot (1.59 + 1.56\Omega)\right] \cdot (m=1.95) \cdot \frac{k_{NI}=1}{k_{NU}=1} \cdot \frac{1}{C=4} = 2.32\,\Omega$$

$$X_{3.St} = \max. = 2.5\Omega$$

B2) Earth factor k, angle φ

$$\underline{k} = \left(\frac{\underline{Z_0}}{\underline{Z_1}} - 1\right) / 3 \qquad \underline{Z_0} = \underline{Z_1} + 3\,\underline{Z_E}$$

$$|\underline{k}| = 0.52$$

$$\varphi = \arctan\frac{X}{R} \approx 65°$$

C) Direction
LS or SS
Set LS
D) Time element

$Z < Z_{1st\ stage}$	Fast time
$Z_{1st\ stage} < Z < Z_{2nd\ stage}$	I. Time delay (here 1s!*)
$Z_{2nd\ stage} < Z < Z_{3rd\ stage}$	II. Time delay (here 2s!*)
$Z > Z_{3rd\ stage}$	III. Time delay (here 2.5s!*)

* - Long switch-off times have been deliberately set to make the difference between the various time delays audible to the participants.

3.3.1.2 Experiments

Network model: Radiating network type B

a) 1-, 2-, and 3-pole short circuit at selected short-circuit locations within the 1st stage (circular characteristic curve)
b) Short circuit at the boundary of the 1st stage
c) Short circuit in the first stage with arc (circular/elliptical characteristic curve)
d) Failure of the subordinate circuit breaker / reserve protection

3.3.2 Tests with the KU relay and the SD 34A distance relay

Network model: Radiating network type B

a) Short circuit in the 1st zone with and without successful arc extinguishing
b) Short circuit in the second zone with and without successful arc extinguishing

3.3.3 Tests with the SIPROTEC 7SA511 distance relay

3.3.3.1 Tests in parallel line

Network model: Parallel line

a) Short circuit within the line
b) Short circuit behind the line with failure of the subordinate circuit breaker

3.3.3.2 Test in ring network

Network model: Ring network

a) Fault detection in ring network
b) Failure of the subordinate circuit breaker
c) Analysis of the fault record on the PC